Case Study 7: Advanced Numeric Pattern Generator

1. Problem Statement

Generate a symmetric hollow diamond pattern for a user-provided odd width.

2. Objectives

Use nested loops and conditions.

Generate spaces and symbols in a pattern.

Validate the input dimension.

3. Assumptions

The width must be an odd integer of at least 3.

The diamond uses asterisks.

The inside of the diamond is hollow except at the top and bottom.

4. Algorithm

1. Start.

2. Input width.

3. If width is even or less than 3, display an error.

4. Generate the upper half including the middle row using nested loops.

5. Generate the lower half in reverse order.

6. Stop.

5. Flowchart (text form)

6. Pseudocode

```
START
INPUT width
IF width < 3 OR width is even: DISPLAY error and STOP
n = width // 2
FOR i from 0 to n: print leading spaces; print border stars and inner spaces FOR i
from n-1 down to 0: print leading spaces; print border stars and inner spaces STOP
```

7. Complete Python Source Code

```
width = int(input("Enter an odd width (3 or more): "))

if width < 3 or width % 2 == 0:
 print("Invalid width. Enter an odd number greater than or equal to 3.")
else:
middle = width // 2

 for i in range(middle + 1):
spaces = middle - i




 print(" " * spaces, end="")

 if i == 0:
 print("*")
 else:
inside_spaces = 2 * i - 1
```

```
  print("*" + " " * inside_spaces + "*")
```

```
   for i in range(middle - 1, -1, -1):
```

spaces = middle - i

```
  print(" " * spaces, end="")
```

```
   if i == 0:
   print("*")
   else:
```

inside_spaces = 2 * i - 1

```
  print("*" + " " * inside_spaces + "*")
```

8. Test Case and Expected Output

Input: 5.

```
* * * * * * * *
```

9. Conclusion

The program validates the width and uses nested loops and string repetition to print a hollow diamond.